Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item →

Instructions

1. Which of the following steps is typically the first in the machine learning analytic process for cyber analysts? 1 / 1 point

- Evaluation
- Data collection
- Feature engineering
- Model deployment

Nice work

Correct! Data collection is usually the first step in the machine learning analytic process as it forms the foundation for all subsequent steps.

2. In the context of machine learning, which of the following is a key objective of linear regression? 1 / 1 point

- To cluster data points into distinct groups based on similarities.
- To classify data points into predefined categories.
- To find the best-fitting line that minimizes the difference between predicted and actual values.
- To reduce the dimensionality of the data by projecting it onto a lower-dimensional space.

Nice work

Correct! Linear regression aims to find a line that best represents the relationship between input features and the target variable by minimizing the error between predicted and actual values.

3. A cybersecurity analyst is using an algorithm to group similar types of cyber threats without relying on labeled outcomes. What type of machine learning is this analyst using? 1 / 1 point

- Reinforcement learning
- Unsupervised machine learning
- Supervised machine learning
- Semi-supervised learning

Nice work

Correct! Unsupervised machine learning is used to cluster or group similar data points without needing labeled outcomes.

4. Which of the following unsupervised machine learning algorithms is commonly used for reducing the dimensionality of large datasets? 1 / 1 point

- Principal Component Analysis (PCA)
- Naive Bayes
- K-means Clustering
- Linear Regression

Nice work

Correct! PCA is an unsupervised machine learning algorithm used to reduce the dimensionality of data while retaining as much variance as possible.

5. A cybersecurity team is working on a project where they need to detect new types of phishing websites based on labeled training data. Which type of machine learning algorithm would be most appropriate for this task? 1 / 1 point

- Supervised machine learning
- Reinforcement learning
- Semi-supervised learning
- Unsupervised machine learning

Nice work

Correct! Supervised machine learning algorithms are suitable for tasks where labeled training data is available for detecting phishing websites.